



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

TSAPI KOUDJOU Alex Cabrel
11/01/2026

<https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone.git>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

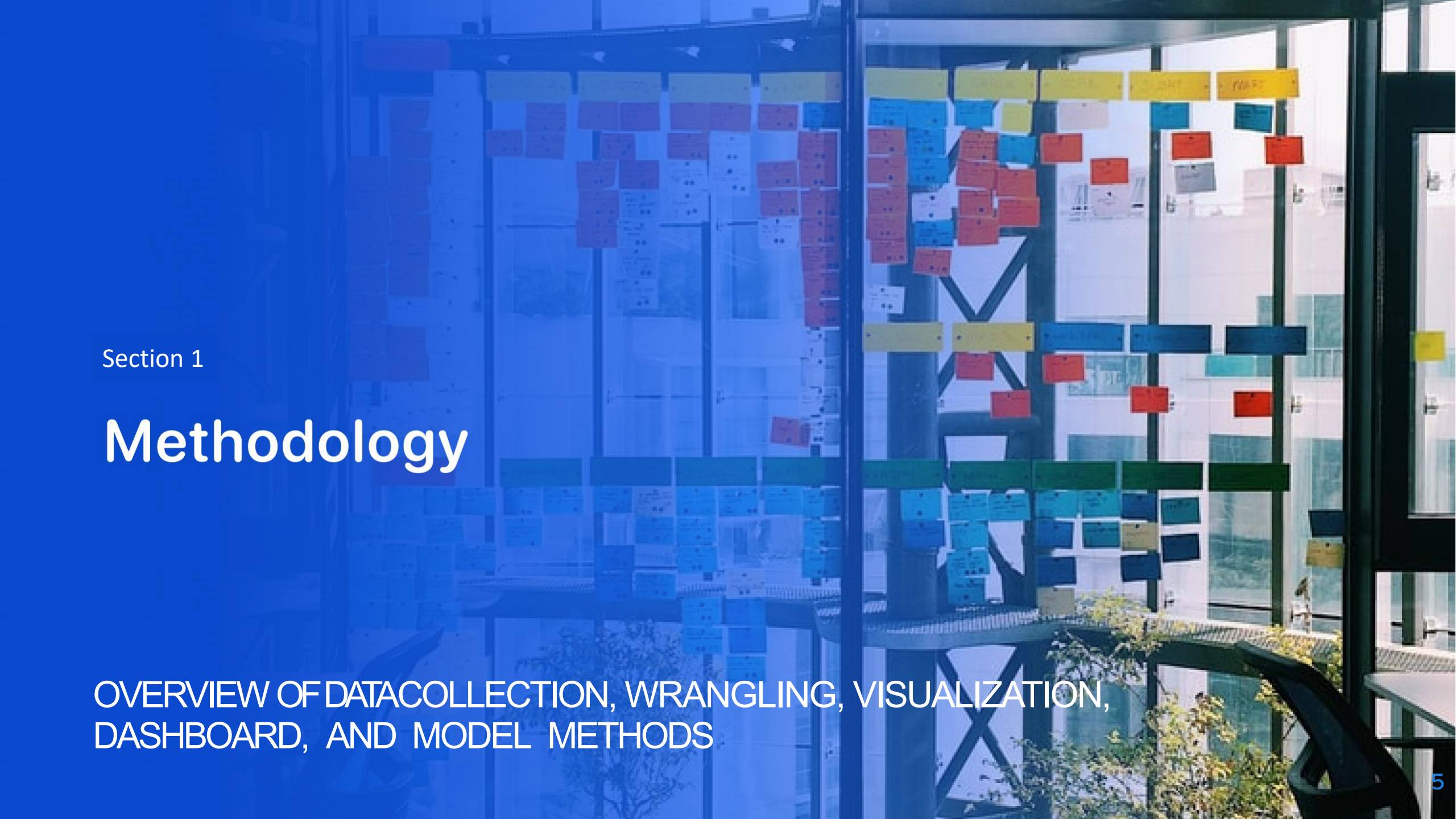
Executive Summary

- Collected data from the public SpaceX API and the SpaceX Wikipedia page. Created a target label, **class**, to identify successful landings. Performed exploratory data analysis using SQL queries, data visualizations, Folium maps, and interactive dashboards. Selected relevant features for modeling, converted categorical variables to binary format using one-hot encoding, and standardized the data. Applied GridSearchCV to identify optimal hyperparameters for machine learning models and visualized the accuracy scores for each model.
- Developed four machine learning models: Logistic Regression, Support Vector Machine, Decision Tree Classifier, and K-Nearest Neighbors. All models achieved similar performance, with an accuracy of approximately 83.33%. Each model tended to overpredict successful landings, indicating potential class imbalance or limited data. Additional data would be beneficial to improve model accuracy and reliability.

Introduction

- **Background**
- The commercial space industry has entered a new competitive era.
- SpaceX offers industry-leading launch pricing (\$62 million compared to \$165 million USD).
- This cost advantage is largely driven by the ability to recover and reuse the first stage of the rocket.
- SpaceY aims to compete directly with SpaceX in the commercial launch market.
- **Problem Statement**
- SpaceY has tasked our team with developing a machine learning model to predict the likelihood of a successful Stage 1 rocket recovery.





Section 1

Methodology

OVERVIEW OF DATA COLLECTION, WRANGLING, VISUALIZATION,
DASHBOARD, AND MODEL METHODS

Methodology

- **Data Collection:** Combined data from the SpaceX public API and the SpaceX Wikipedia page.
- **Data Wrangling:** Labeled landings as successful when recovery occurred and unsuccessful otherwise.
- **Exploratory Data Analysis (EDA):** Conducted analysis using SQL queries and data visualizations to identify patterns and trends.
- **Interactive Visual Analytics:** Built interactive maps and dashboards using Folium and Plotly Dash.
- **Predictive Modeling:** Performed classification-based predictive analysis to estimate landing success.
- **Model Tuning:** Optimized model performance using GridSearchCV for hyperparameter selection.

Data Collection

- **Data Collection Overview**

- Data was collected using two sources: the **SpaceX public API** and **web scraping** from SpaceX's Wikipedia launch tables.
- The next slides illustrate the data collection workflows for the API and web scraping processes.

- **SpaceX API Data**

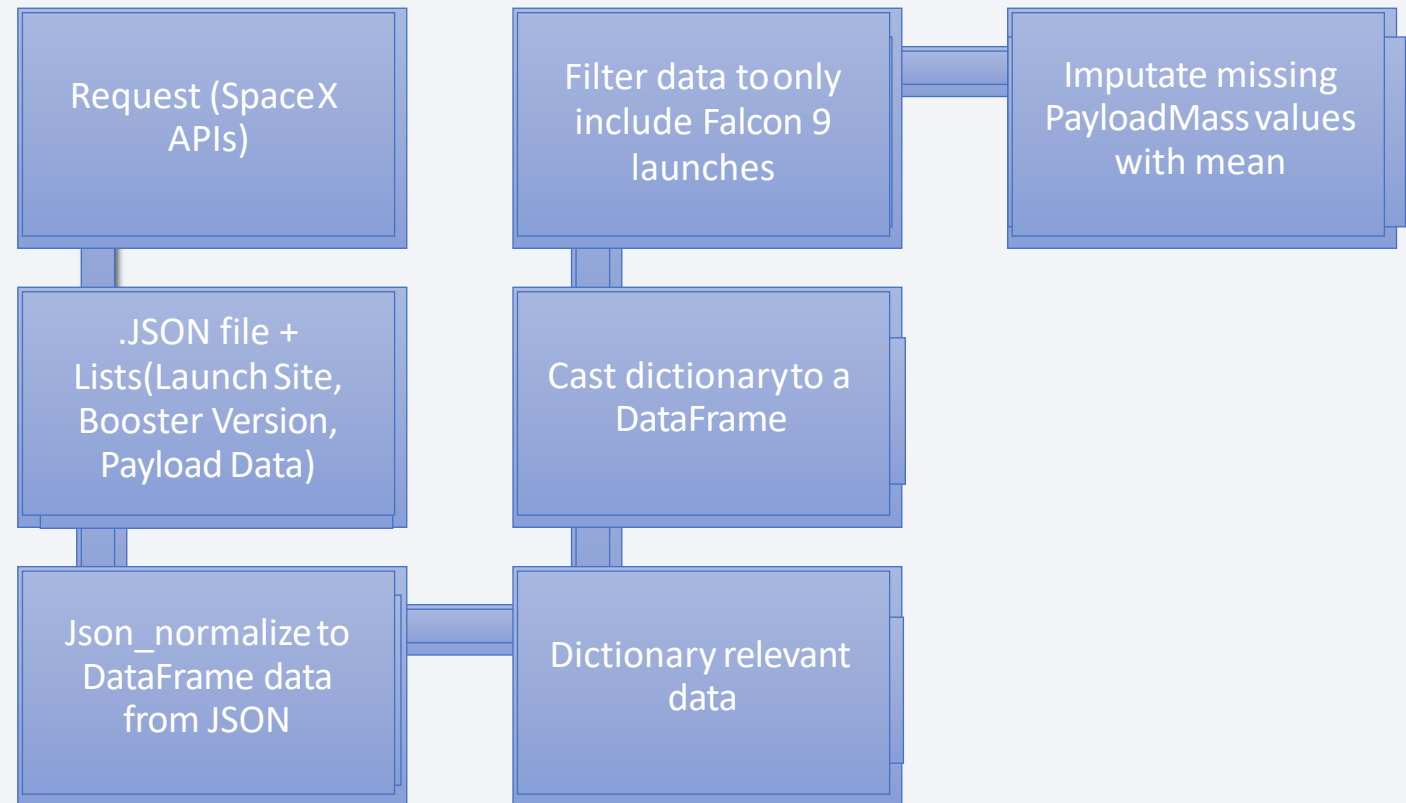
- Flight number, date, booster version, payload mass, orbit
- Launch site, landing outcome, reuse indicators (flights, grid fins, legs, reused)
- Landing pad, block, reuse count, serial number
- Launch location coordinates (latitude, longitude)

- **Wikipedia Web-Scraped Data**

- Flight number, launch site, payload, payload mass
- Orbit, customer, launch outcome
- Booster version, booster landing status
- Date and time

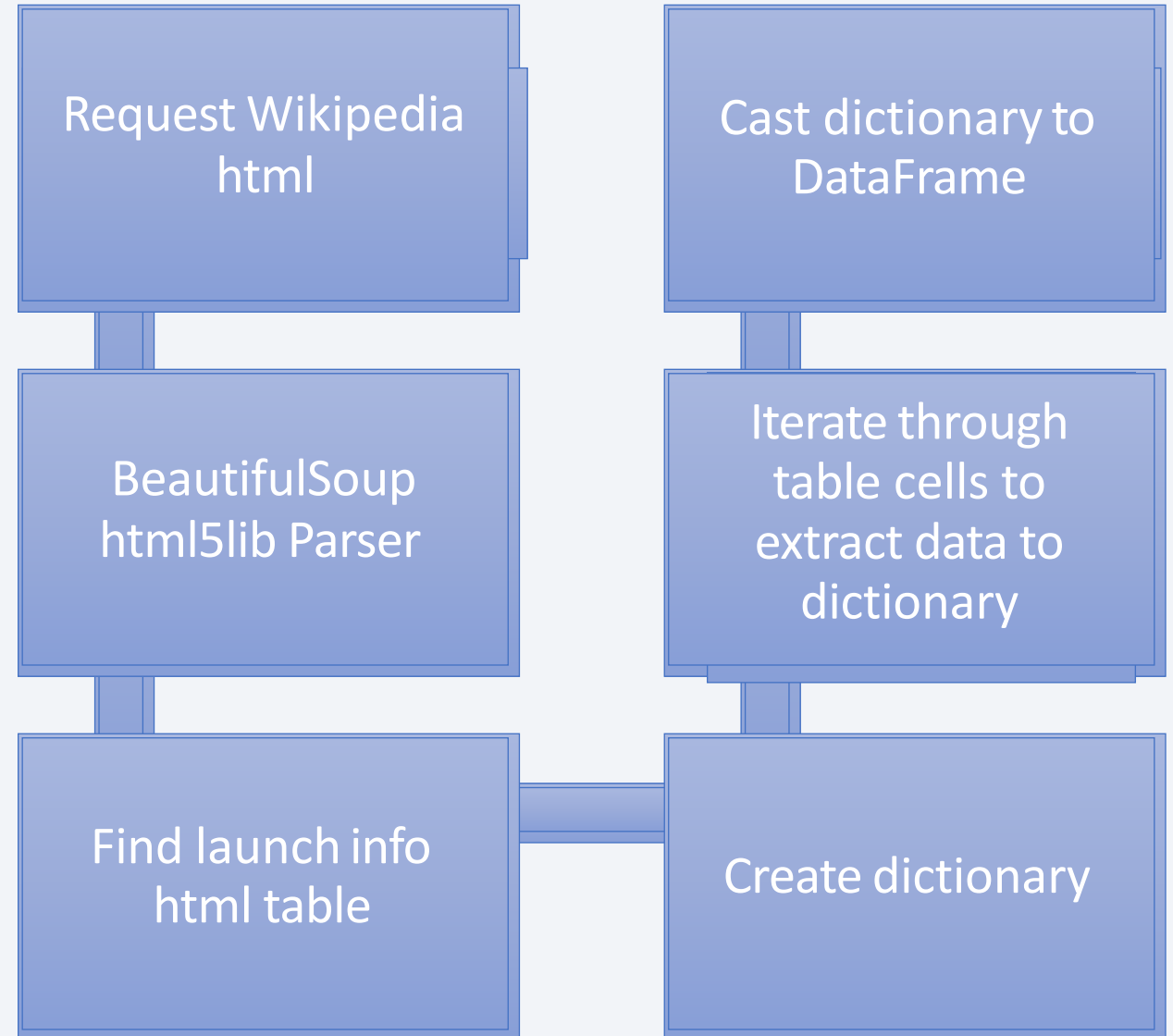
Data Collection – SpaceX API

- <https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone/blob/main/Data%20Collection%20Api.ipynb>



Data Collection - Scraping

<https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone/blob/main/Data%20Collection%20with%20Web%20Scraping.ipynb>



Data Wrangling

- Created a binary training label for landing outcomes (**success = 1, failure = 0**).
- The **Outcome** field contains two parts: **Mission Outcome** and **Landing Location**.
- Added a new target column, **class**, set to **1** when **Mission Outcome = True**, and **0** otherwise.
- **Value Mapping**
- **1 (Successful):** True ASDS, True RTLS, True Ocean
- **0 (Unsuccessful):** None None, False ASDS, None ASDS, False Ocean, False RTLS
- <https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone/blob/main/Data%20wrangling.ipynb>

EDA with Data Visualization

- Exploratory data analysis was conducted on the following variables: **Flight Number, Payload Mass, Launch Site, Orbit, Class, and Year.**
- **Visualizations Used**
- Flight Number vs. Payload Mass
- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Orbit vs. Success Rate
- Flight Number vs. Orbit
- Payload Mass vs. Orbit
- Yearly Success Trend
- Scatter plots, line charts, and bar charts were used to examine relationships between variables. These visualizations helped identify meaningful patterns and correlations to determine which features were suitable for training the machine learning models.

Github URL:

<https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone/blob/main/EDA%20with%20Visualization.ipynb>

EDA with SQL

- Exploratory Data Analysis performed on variables Flight Number, Payload Mass, Launch Site, Orbit, Class and Year.
- Plots Used:
 - Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload Mass vs. Launch Site, Orbit vs. Success Rate, Flight Number vs. Orbit, Payload vs Orbit, and Success Yearly Trend
 - Scatter plots, line charts, and bar plots were used to compare relationships between variables to
 - decide if a relationship exists so that they could be used in training the machine learning model

Github URL:

<https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone/blob/main/EDA%20with%20SQL.ipynb>

Build an Interactive Map with Folium

- Folium maps were used to visualize **launch sites, successful and unsuccessful landings, and proximity to key infrastructure**, including **railways, highways, coastlines, and nearby cities**.
- These visualizations help explain why launch sites are located in specific areas and illustrate how landing success relates to geographic location.
- **GitHub Repository:** <https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone/blob/main/Interactive%20Visual%20Analytics%20with%20Folium.ipynb>

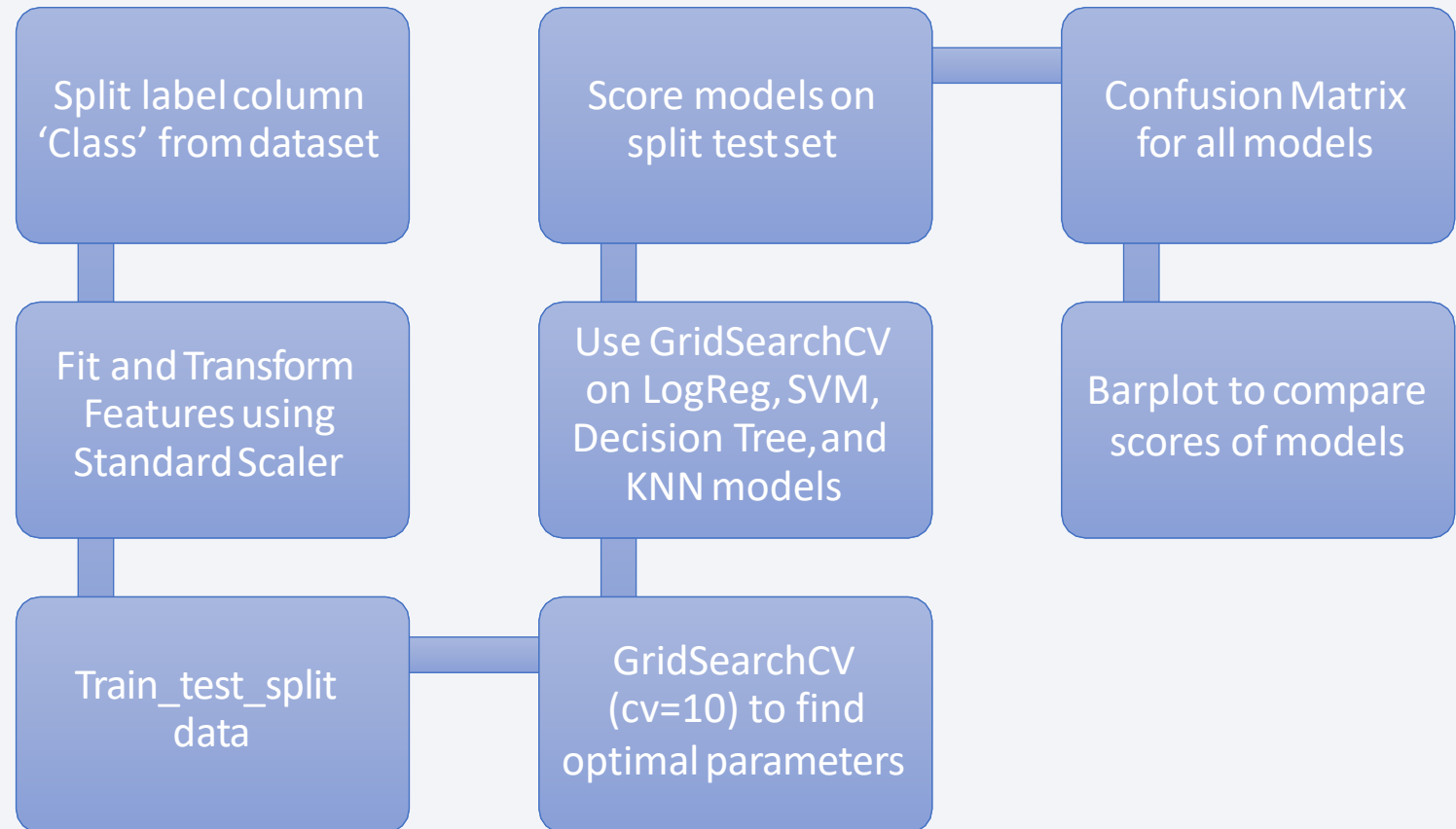
Build a Dashboard with Plotly Dash

- The dashboard includes an **interactive pie chart** and a **scatter plot**.
- **Pie Chart:**
Displays the distribution of successful landings across all launch sites or can be filtered to show success rates for an individual launch site.
- **Scatter Plot:**
Allows selection of all launch sites or a specific site and includes a payload mass slider ranging from **0 to 10,000 kg**.
- The pie chart highlights launch site success rates, while the scatter plot illustrates how landing success varies by **launch site, payload mass, and booster version category**.
- **GitHub Repository:** https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone/blob/main/spacex_dash_app.py

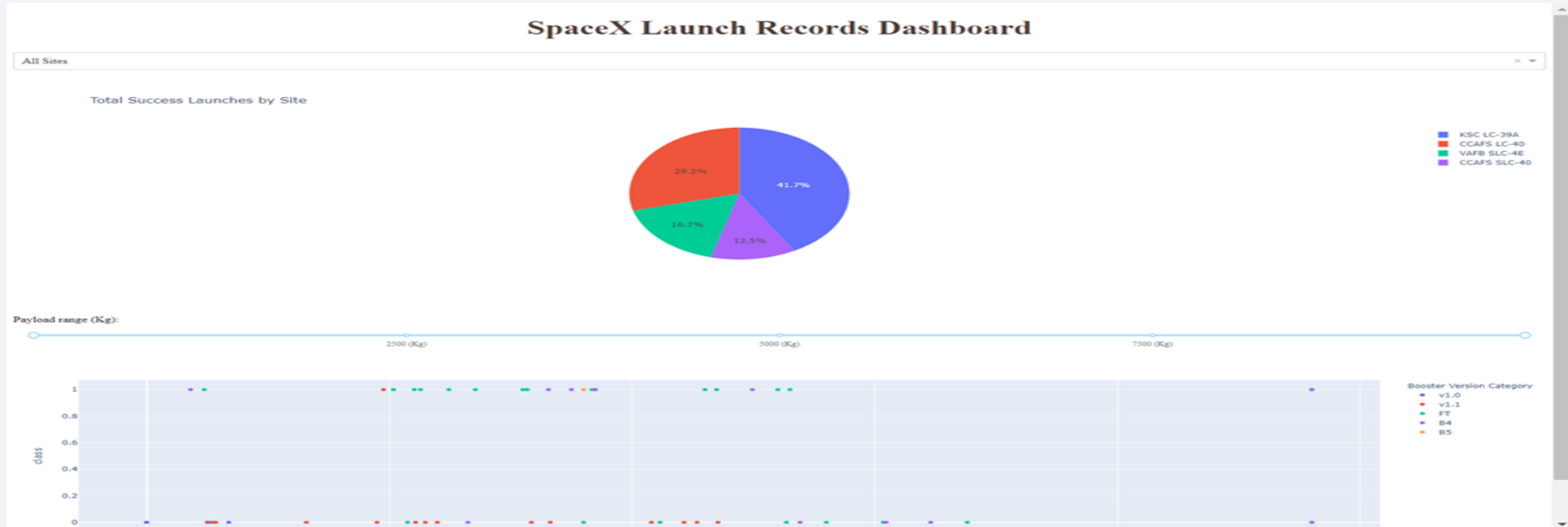
Predictive Analysis (Classification)

Github URL:

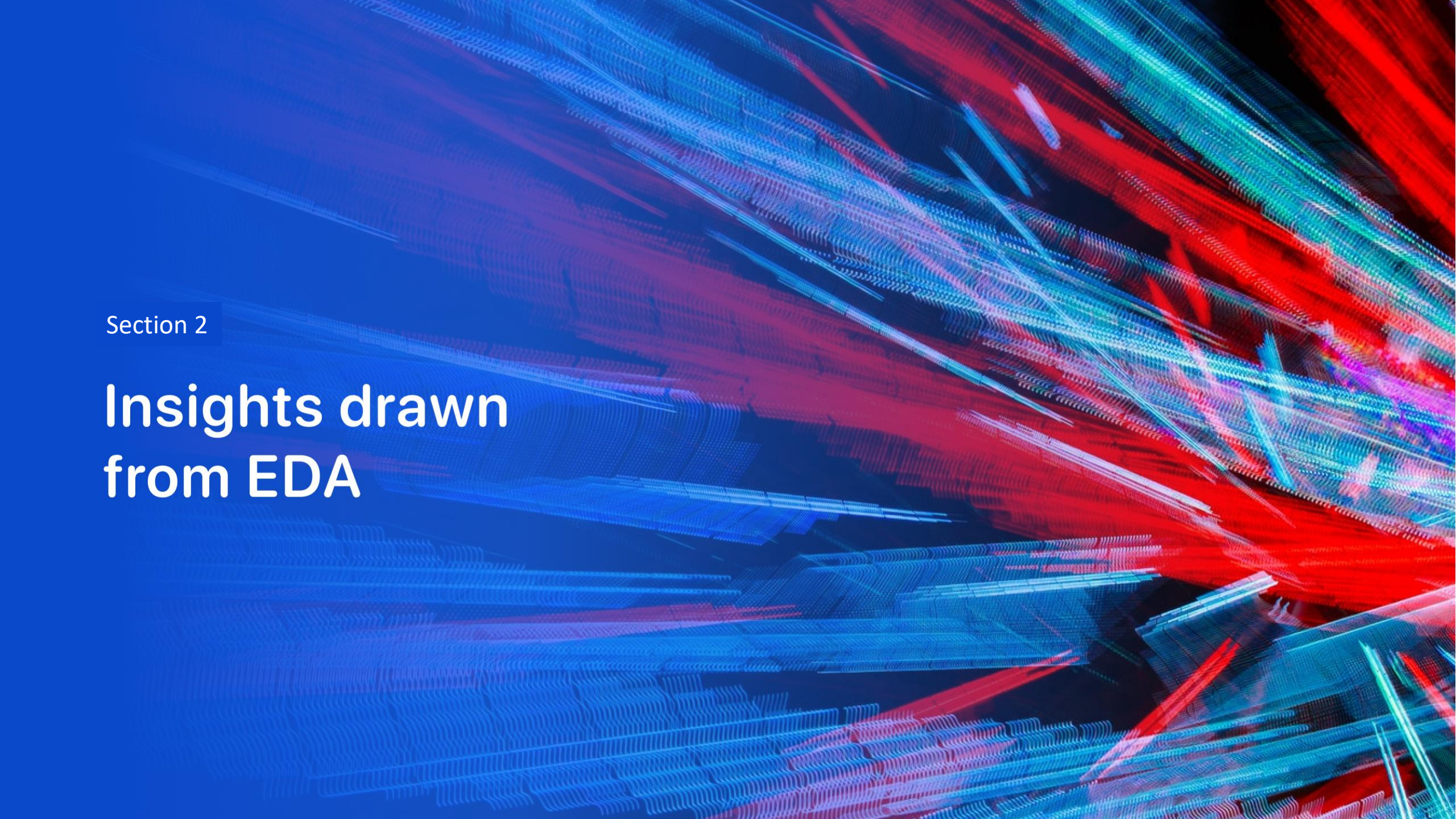
<https://github.com/alex-05-20/IBM-Data-Science-Applied-Capstone/blob/main/Machine%20Learning%20Prediction.ipynb>



Results



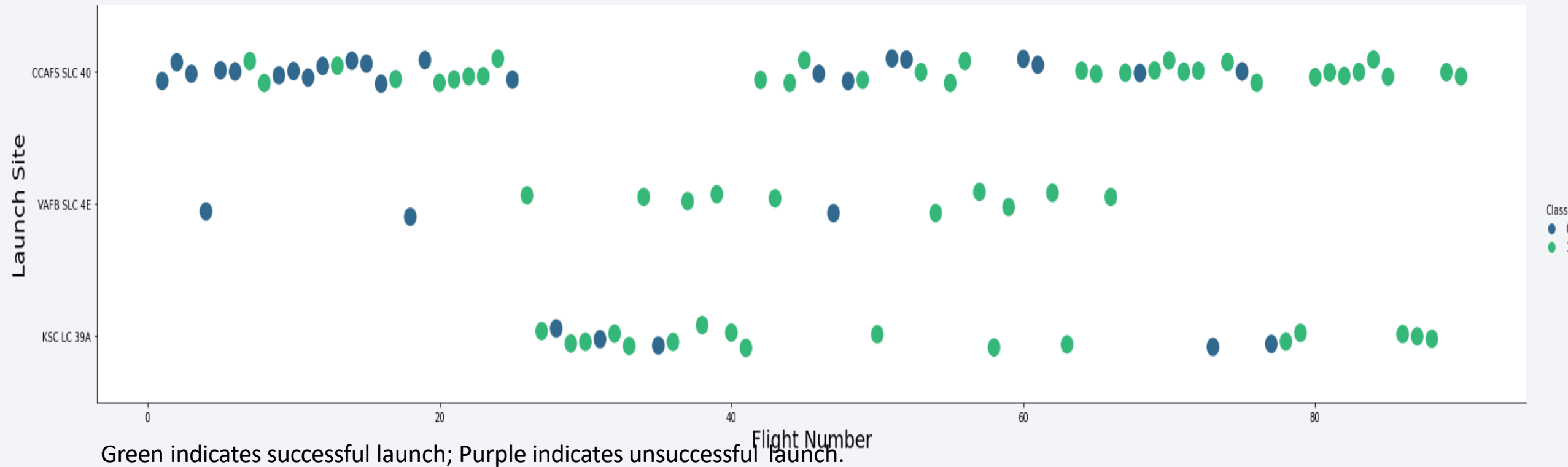
- This slide provides a preview of the **Plotly dashboard**. The following slides present the results of **exploratory data analysis (EDA)** using visualizations and SQL, **interactive geographic analysis** with Folium, and finally the **machine learning model results**, which achieve an accuracy of approximately **83%**.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the upper right quadrant. The overall effect is dynamic and technological.

Section 2

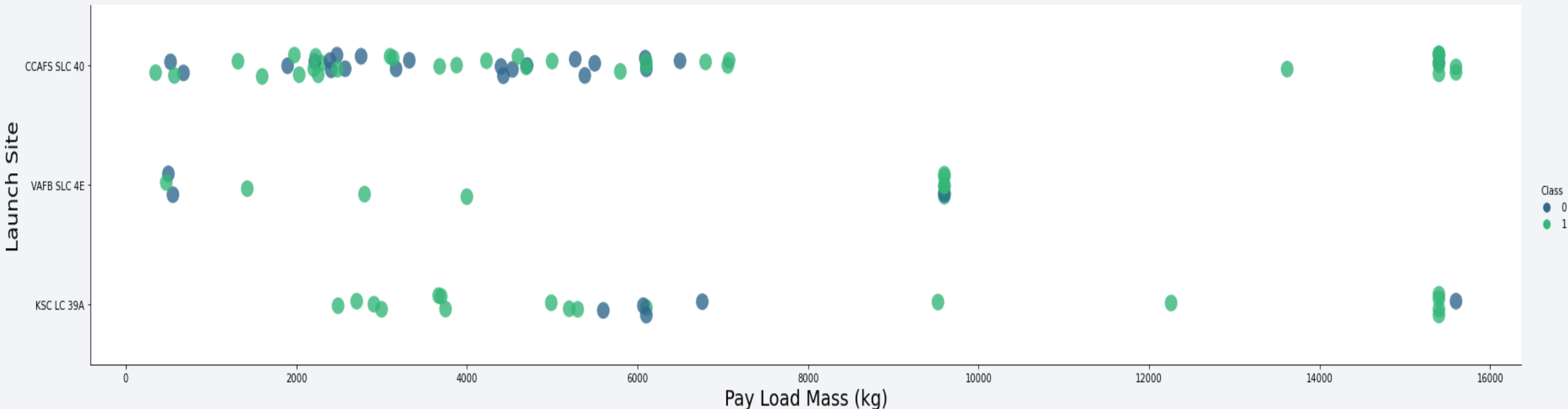
Insights drawn from EDA

Flight Number vs. Launch Site



- The visualization indicates a clear increase in landing success rate over time, as reflected by the flight number. A notable improvement appears around **Flight 20**, suggesting a major operational or technological breakthrough that significantly improved outcomes. Additionally, **CCAFS** emerges as the primary launch site, accounting for the highest launch volume.

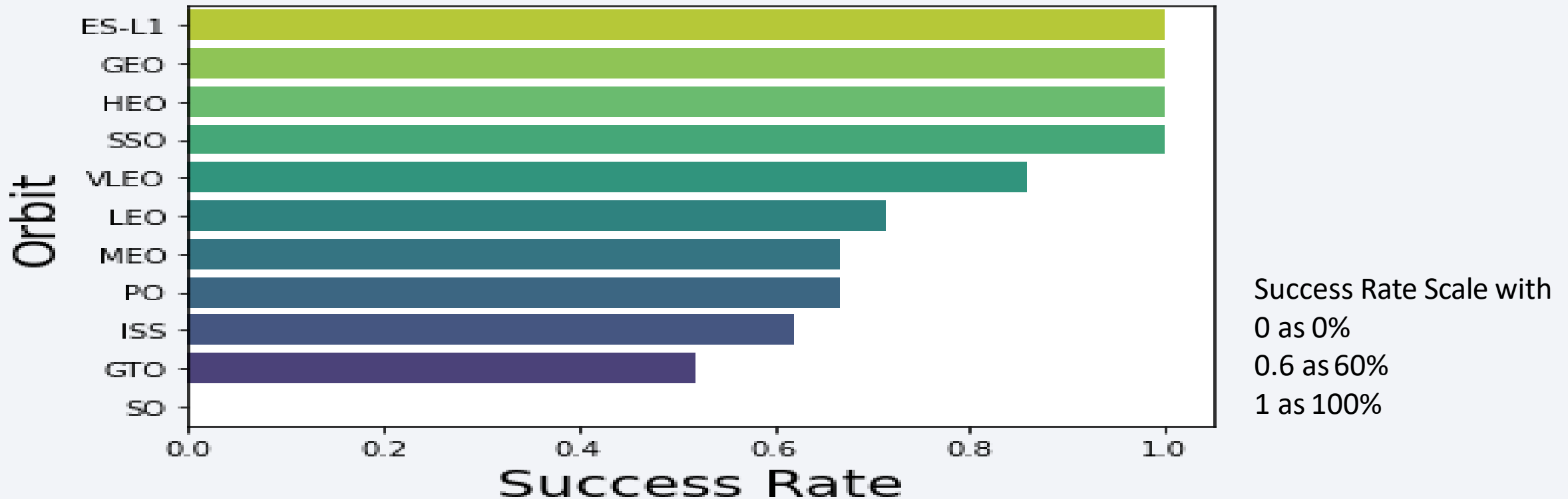
Payload vs. Launch Site



Green indicates successful launch; Purple indicates unsuccessful launch.

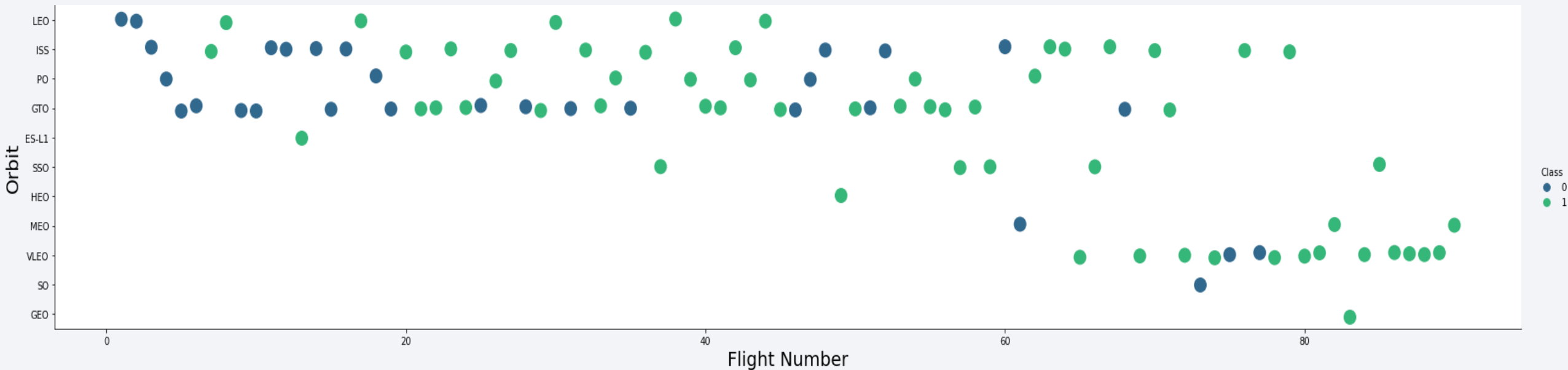
Payload mass is predominantly concentrated between **0 and 6,000 kg**. Additionally, different launch sites tend to support **distinct payload mass ranges**, indicating site-specific mission profiles.

Success Rate vs. Orbit Type



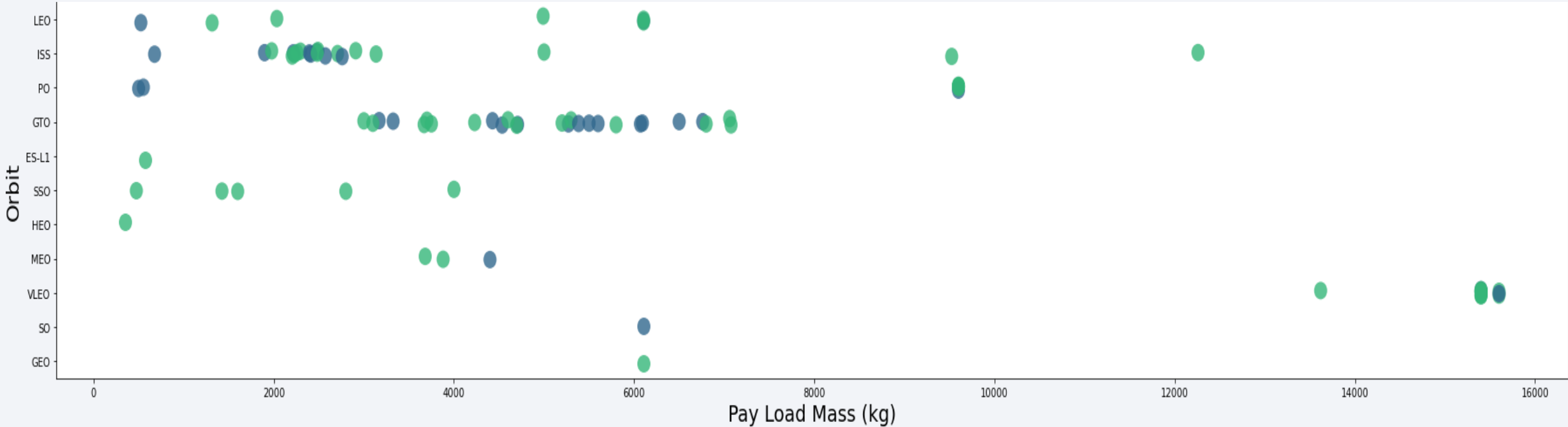
- **ES-L1 (1), GEO (1), HEO (1):** 100% success rate (*very small sample sizes*)
- **SSO (5):** 100% success rate
- **VLEO (14):** Moderate success rate with a reasonable number of attempts
- **SO (1):** 0% success rate (*single attempt*)
- **GTO (27):** Approximately 50% success rate with the **largest sample size**

Flight Number vs. Orbit Type



Green indicates successful launch; Purple indicates unsuccessful launch.

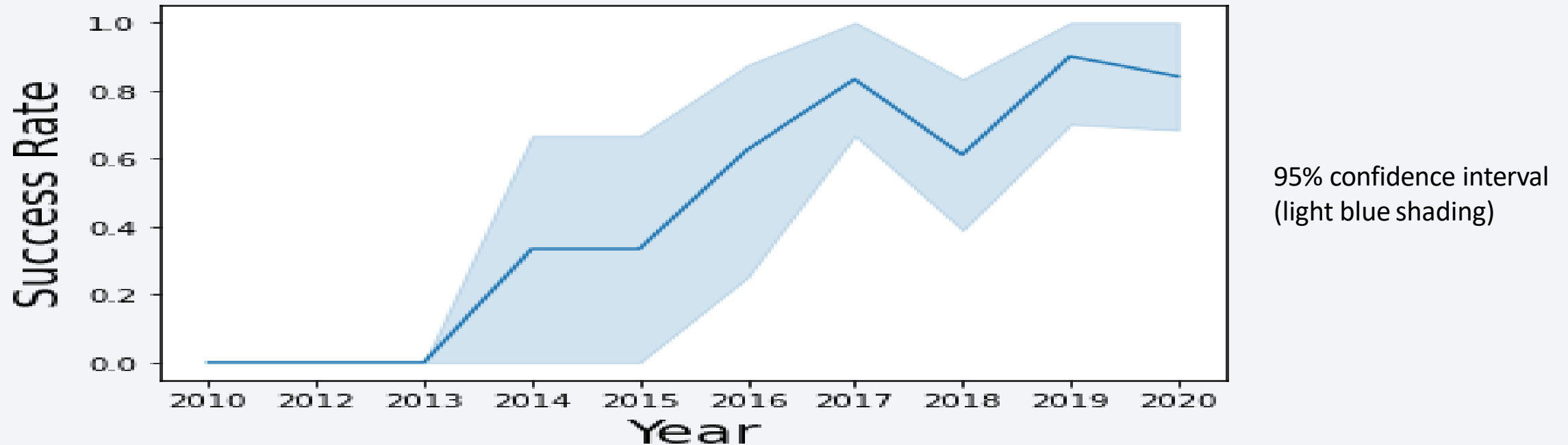
- Launch orbit preferences evolved over time, as indicated by flight number, and launch outcomes appear to correlate with these changes. SpaceX initially focused on **LEO** missions with moderate success, later shifting toward **VLEO** in more recent launches. Overall, performance appears stronger in **lower Earth orbits** and **Sun-synchronous orbits**.



Green indicates successful launch; Purple indicates unsuccessful launch.

- Payload mass shows a clear relationship with orbital type. **LEO** and **SSO** missions generally carry **lower payload masses**, while **VLEO**, another highly successful orbit category, is associated with **higher-end payload mass values**.

Launch Success Yearly Trend



Success generally increases over time since 2013 with a slight dip in 2018

Success in recent years at around 80%

All Launch Site Names

- Queried the database to identify **unique launch site names**.
- Identified inconsistencies caused by **data entry variations** (e.g., *CCAFS SLC-40* vs. *CCAFSSLC-40*), which refer to the same launch site.
- Recognized that **CCAFS LC-40** is an earlier naming convention for the same location.
- After standardization, the data contains **three unique launch sites**:
 - **CCAFS SLC-40**
 - **KSC LC-39A**
 - **VAFB SLC-4E**

```
In [4]: %%sql
        SELECT UNIQUE LAUNCH_SITE
        FROM SPACEXDATASET;

* ibm_db_sa://ftb12020:***@0c77d6f:
Done.
```

Out[4]:

launch_site
CCAFS LC-40
CCAFS SLC-40
CCAFSSLC-40
KSC LC-39A
VAFB SLC-4E

Launch Site Names Begin with 'CCA'

In [5]:

```
%%sql
SELECT *
FROM SPACEXDATASET
WHERE LAUNCH_SITE LIKE 'CCA%'
LIMIT 5;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8l1cg.databases.appdomain.cloud:31198/bludb
Done.
```

Out[5]:

DATE	time__utc_	booster_version	launch_site	payload	payload_mass__kg_	orbit	customer	mission_outcome	landing__outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- First five entries in database with Launch Site name beginning with CCA.

Total Payload Mass

```
%%sql
SELECT SUM(PAYLOAD_MASS_KG_) AS SUM_PAYLOAD_MASS_KG
FROM SPACEXDATASET
WHERE CUSTOMER = 'NASA (CRS)';

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86
Done.
```

sum_payload_mass_kg

45596

- This query calculates the **total payload mass (kg)** for missions where **NASA** was the customer. **CRS** stands for **Commercial Resupply Services**, indicating that these payloads were delivered to the **International Space Station (ISS)**.

Average Payload Mass by F9 v1.1

```
%%sql
SELECT AVG(PAYLOAD_MASS_KG_) AS AVG_PAYLOAD_MASS_KG
FROM SPACEXDATASET
WHERE booster_version = 'F9 v1.1'
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86
Done.
```

avg_payload_mass_kg
2928

- This query calculates the **average payload mass** for launches using the **Falcon 9 v1.1** booster.
The results show that the **average payload mass for F9 v1.1** falls on the **lower end of the overall payload mass range**.

First Successful Ground Landing Date

```
%%sql
SELECT MIN(DATE) AS FIRST_SUCCESS
FROM SPACEXDATASET
WHERE landing_outcome = 'Success (ground pad)';

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81
Done.
```

first_success
2015-12-22

- This query returns the first successful ground pad landing date.
- First ground pad landing wasn't until the end of 2015.
- Successful landings in general appear starting 2014.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql
SELECT booster_version
FROM SPACEXDATASET
WHERE landing__outcome = 'Success (drone ship)' AND payload_mass__kg_ BETWEEN 4001 AND 5999;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90108kqb1od8l1cg.database
Done.
```

booster_version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- This query returns the four booster versions that had successful drone ship landings and a payload mass between 4000 and 6000 noninclusively.

Total Number of Successful and Failure Mission Outcomes

```
%%sql
SELECT mission_outcome, COUNT(*) AS no_outcome
FROM SPACEXDATASET
GROUP BY mission_outcome;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-1
Done.
```

mission_outcome	no_outcome
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

- This query returns the **count of each mission outcome**.
The results show that **SpaceX achieves its primary mission objective in nearly 99% of launches**, indicating a very high mission success rate.
- This suggests that most **landing failures are intentional**, often occurring when recovery is not required for mission success. Additionally, one launch has an **unclear payload status**, and **one mission failed during flight**.

Boosters Carried Maximum Payload

```
%%sql
SELECT booster_version, PAYLOAD_MASS__KG_
FROM SPACEXDATASET
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXDATASET);

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1
Done.
```

booster_version	payload_mass__kg_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

- his query identifies the **booster versions** that carried the **maximum payload mass of 15,600 kg**. All of these boosters belong to the **Falcon 9 Block 5 (B10xx.x)** family and are highly similar in design.
- This pattern suggests a strong relationship between **booster version** and **payload capacity**.

2015 Launch Records

```
%%sql
SELECT MONTHNAME(DATE) AS MONTH, landing_outcome, booster_version, PAYLOAD_MASS_KG_, launch_site
FROM SPACEXDATASET
WHERE landing_outcome = 'Failure (drone ship)' AND YEAR(DATE) = 2015;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.app
Done.
```

MONTH	landing_outcome	booster_version	payload_mass_kg_	launch_site
January	Failure (drone ship)	F9 v1.1 B1012	2395	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	1898	CCAFS LC-40

- This query returns the Month, Landing Outcome, Booster Version, Payload Mass (kg), and Launch site of 2015 launches where stage 1 failed to land on a drone ship.
- There were two such occurrences.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT landing_outcome, COUNT(*) AS no_outcome
FROM SPACEXDATASET
WHERE landing_outcome LIKE 'Success%' AND DATE BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY landing_outcome
ORDER BY no_outcome DESC;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lce
Done.
```

landing_outcome	no_outcome
Success (drone ship)	5
Success (ground pad)	3

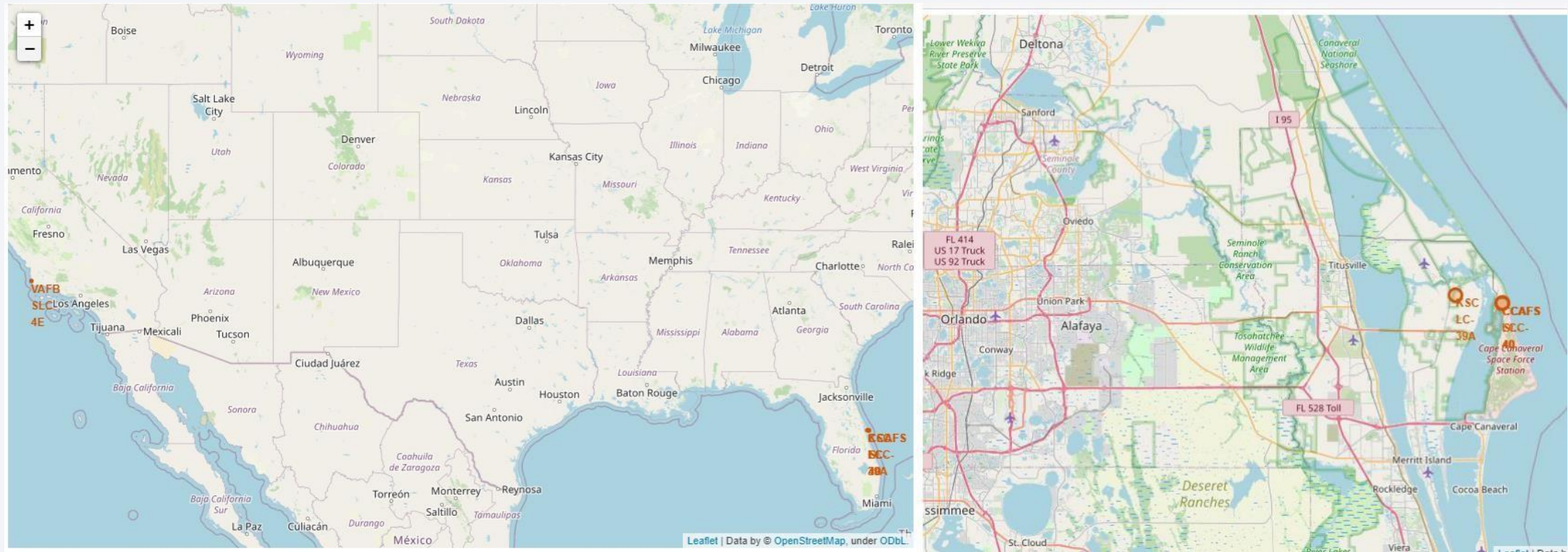
- This query returns all **successful landings** between **June 4, 2010** and **March 20, 2017** (inclusive). During this period, there were **two types of successful landing outcomes: drone ship landings and ground pad landings**.
- In total, **8 successful landings** occurred within this timeframe.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

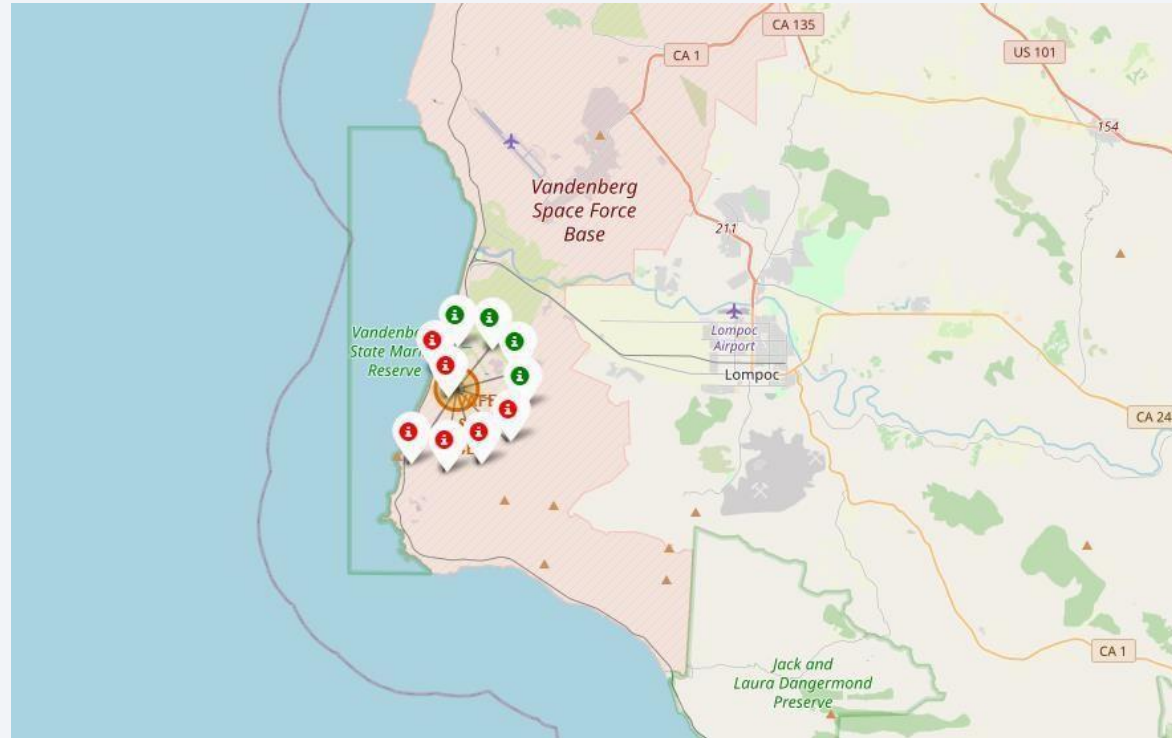
Launch Sites Proximities Analysis

Launch Site Locations



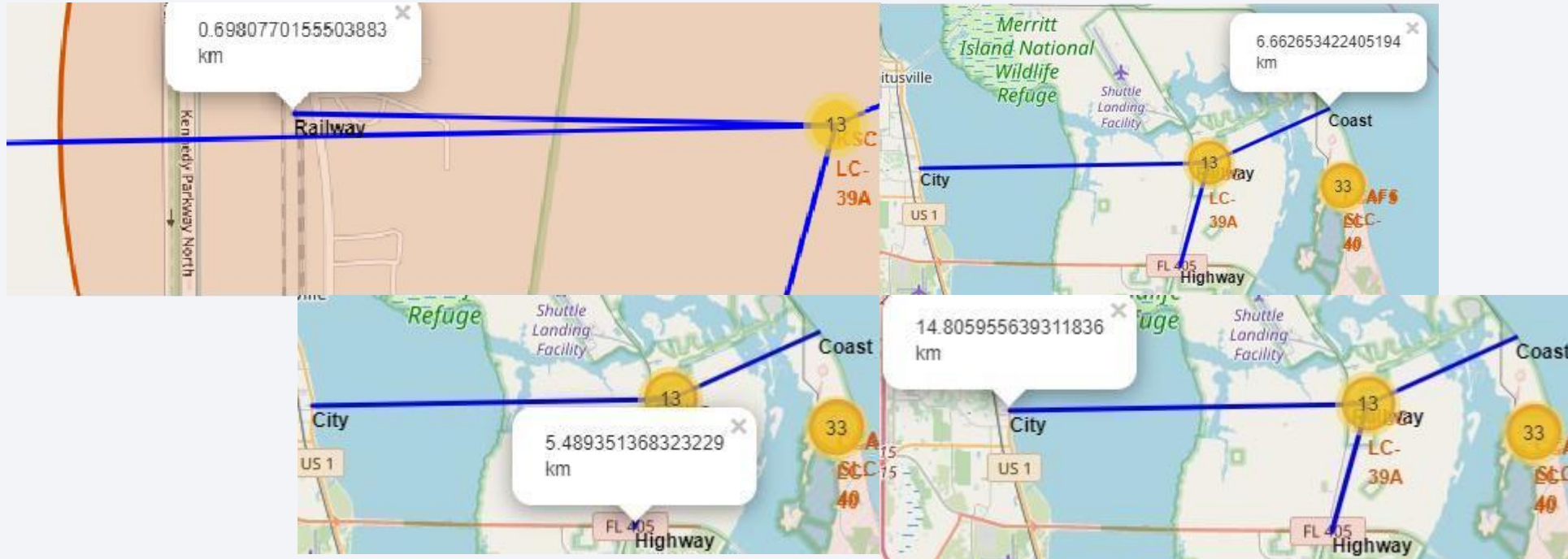
The left map shows all launch sites relative US map. The right map shows the two Florida launch sites since they are very close to each other. All launch sites are near the ocean.

Color-Coded Launch Markers

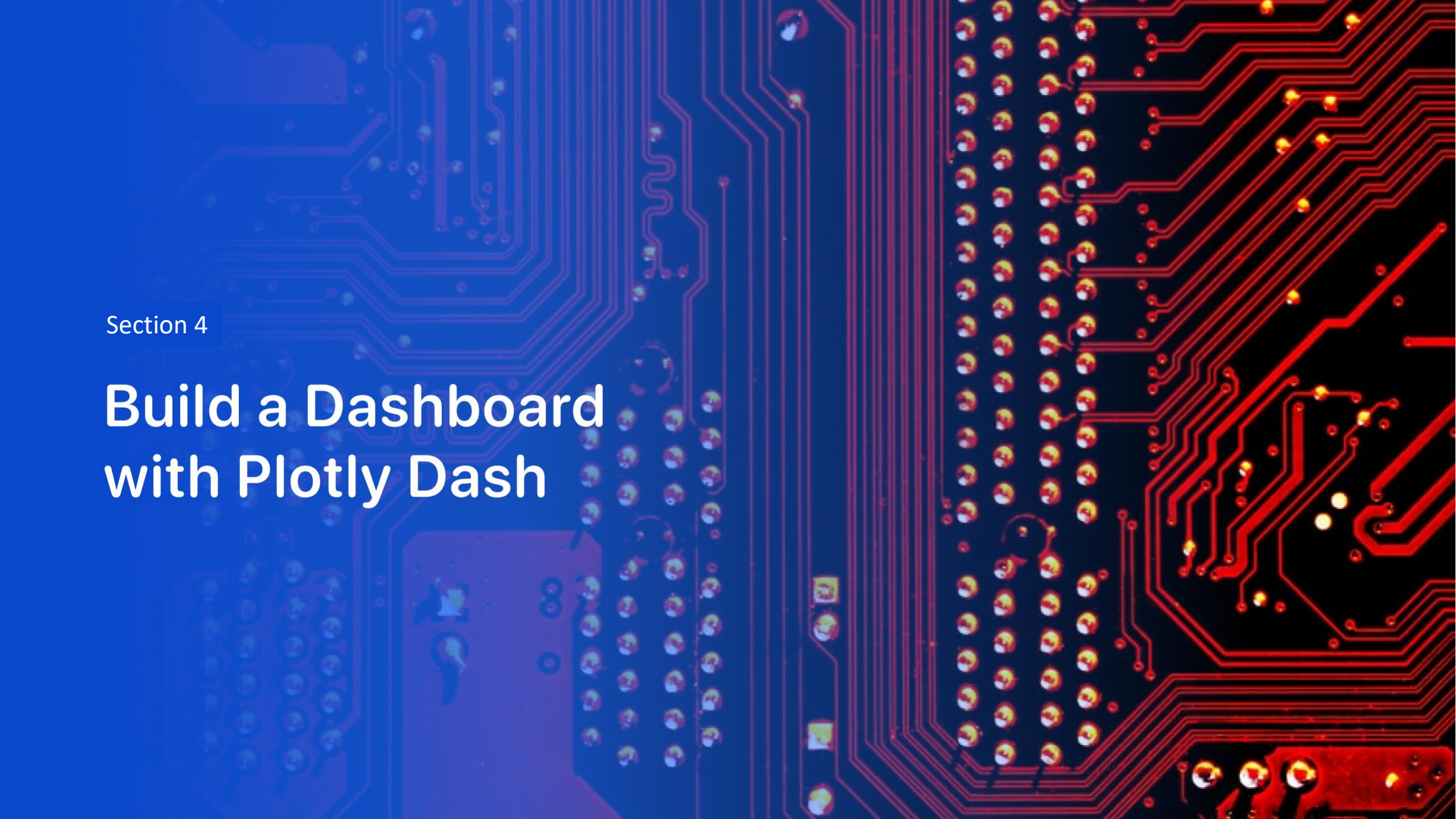


Clusters on Folium map can be clicked on to display each successful landing (green icon) and failed landing (red icon). In this example VAFB SLC-4E shows 4 successful landings and 6 failed landings.

Key Location Proximities



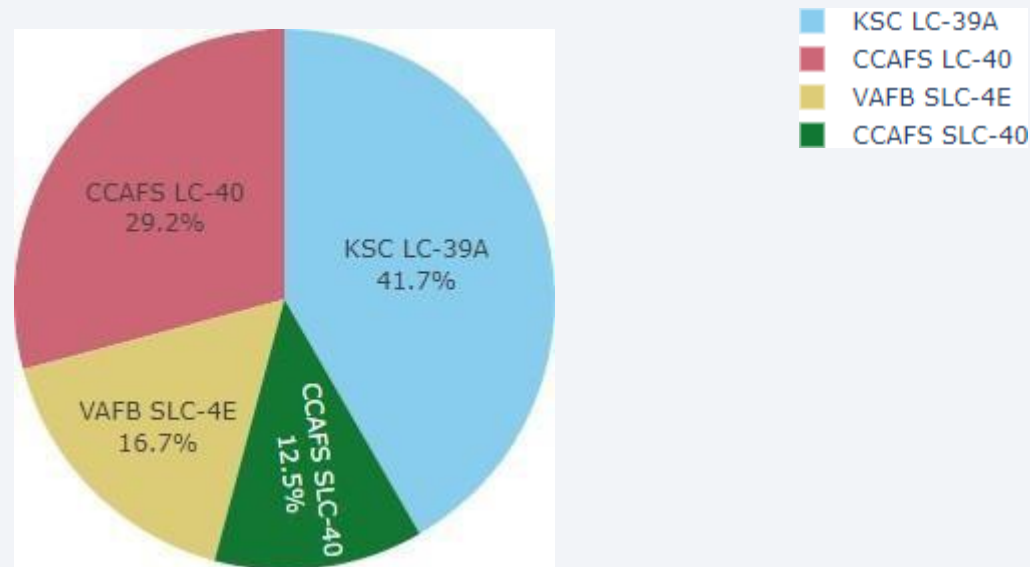
Using KSC LC-39A as an example, launch sites are very close to railways for large part and supply transportation. Launch sites are close to highways for human and supply transport. Launch sites are also close to coasts and relatively far from cities so that launch failures can land in the sea to avoid rockets falling on densely populated areas.



Section 4

Build a Dashboard with Plotly Dash

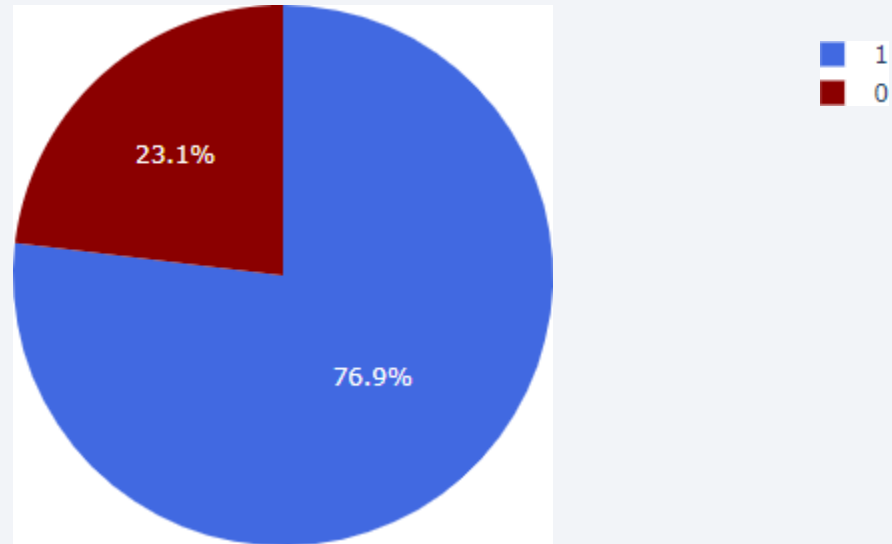
Successful Launches Across Launch Sites



This is the distribution of successful landings across all launch sites. CCAFS LC-40 is the old name of CCAFS SLC-40 so CCAFS and KSC have the same amount of successful landings, but a majority of the successful landings were performed before the name change. VAFB has the smallest share of successful landings. This may be due to smaller sample and increase in difficulty of launching in the west coast. [39](#)

Highest Success Rate Launch Site

KSC LC-39A Success Rate (blue=success)



KSC LC-39A has the highest success rate with 10 successful landings and 3 failed landings.

Payload Mass vs. Success vs. Booster Version Category

Payload range (Kg):



Payload Mass vs. Success vs. Booster Version Category



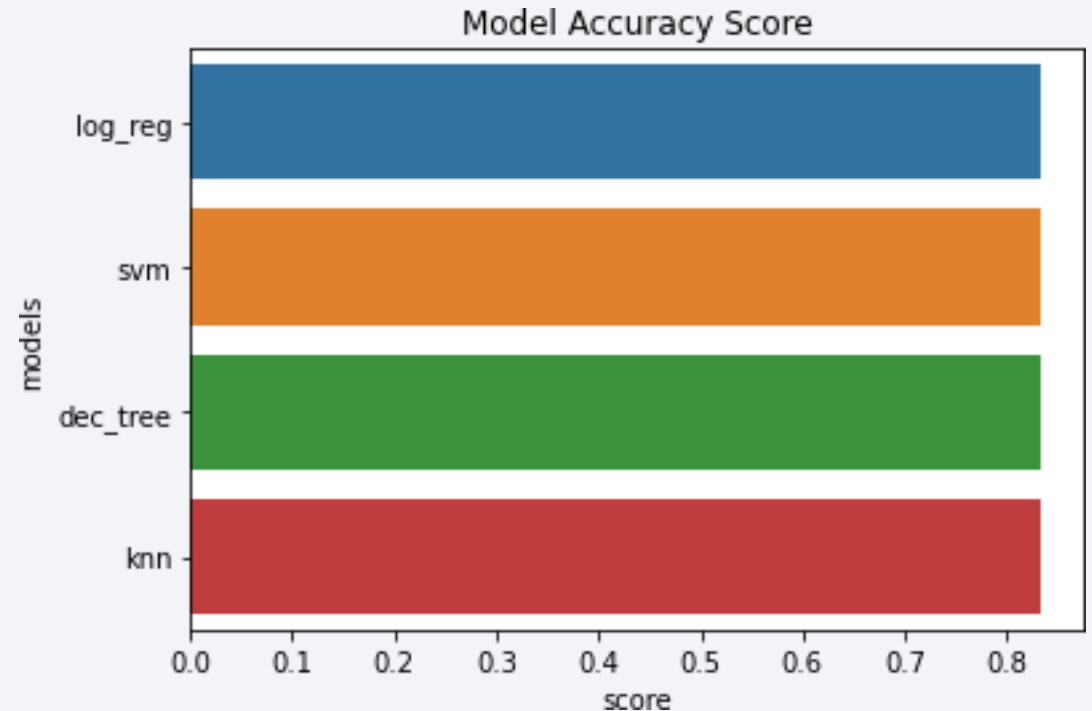
Plotly dashboard has a Payload range selector. However, this is set from 0-10000 instead of the max Payload of 15600. Class indicates 1 for successful landing and 0 for failure. Scatter plot also accounts for booster version category in color and number of launches in point size. In this particular range of 0-6000, interestingly there are two failed landings with payloads of zero kg.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- All models achieved nearly identical test accuracy of **83.33%**. However, the test set is very small (**n = 18**), which can lead to **high variance in performance estimates**, particularly for models such as the **Decision Tree Classifier**, where results may fluctuate across repeated runs.
- Additional data is likely required to **reliably distinguish the best-performing model**.



Confusion Matrix

Because all models performed identically on the test set, the **confusion matrix is the same across all models**.

True Positives: 12 successful landings correctly predicted

True Negatives: 3 unsuccessful landings correctly predicted

False Positives: 3 unsuccessful landings incorrectly predicted as successful

These results indicate that the models tend to **overpredict successful landings**, which may be influenced by class imbalance or limited data.



Correct predictions are on a diagonal from top left to bottom right.

Conclusions

- Developed a **machine learning model** for **SpaceY** to support competitive bidding against SpaceX.
- The model predicts whether **Stage 1 recovery** will be successful, with the potential to save **approximately \$100 million USD per launch**.
- Data was collected from the **public SpaceX API** and **web-scraped from SpaceX's Wikipedia pages**.
- Created labeled datasets and stored them in a **DB2 SQL database**.
- Built **interactive dashboards** to support exploratory analysis and visualization.
- Trained multiple machine learning models, achieving a **best accuracy of ~83%**.
- This model can be used pre-launch to estimate the likelihood of a successful Stage 1 landing and inform **go/no-go launch decisions**.
- Collecting additional data would help **reduce variance**, improve accuracy, and better identify the optimal model.

Appendix

- Special thanks to all my instructors.
- Instructors:
- **Instructors: Rav Ahuja, Alex Aklson, Aije Egwaikhide, Svetlana Levitan, Romeo Kienzler, Polong Lin, Joseph Santarcangelo, Azim Hirjani, Hima Vasudevan, Saishruthi Swaminathan, Saeed Aghabozorgi, Yan Luo**

Thank you!

